Qingteng Zhang, Ph.D., Physicist

X-ray Science Division
Argonne National Laboratory
9700 South Cass Avenue
Argonne, IL 60439

☎: 630-252-6366

Significance Statement

Temperature, Not Structure, Sets Dynamic Arrest in an Elastin-like Protein Polymer

Ozgulbas, Chu, Lavens, Marks, Du, Zuo, Irvin, Ma, Seifert, Lee, Vescovi, Foster, Mateevitsi, Jensen, Akins, Ramanathan, Babnigg and Zhang

Physical Sciences: Applied Physical Sciences; secondary, Biophysics and Computational Biology

Whether a soft material has solidified is usually inferred from its structure: a scattering pattern that has grown enough is read as a gel. We show that this inference can fail. In a designed protein polymer, samples held within a degree of one another develop nearly the same structure, yet only the warmer ones stop moving. Seeing this needed X-ray speckle framed at fifty thousand images per second, because the motion reporting on arrest is millisecond-scale, below what conventional detectors reach. Arrest spreads region by region, largest scales first, and one rescaling of waiting time collapses every sample onto a single pathway whose temperature scale is a fifth of a degree, too sharp for single contacts to set.

118 words. Limit 120.